

PUSHPAK 02

The Future of Intercity & Urban Mobility

Hydrogen-Electric Aircraft

Transforming 100–300 km regional air routes + premium urban mobility with clean, high-endurance hydrogen-electric aviation.

**ROUND: Series A | TARGET:
\$23.8M | DATE: June 2026**



Executive Summary

Transforming Regional & Urban Mobility

Pushpak 02 is building hydrogen-electric aircraft optimized for intercity (100-300km) and premium urban air mobility, solving the range limitations of battery eVTOLS and emissions of conventional helicopters.

300+ km Range

Enables critical intercity routes and premium urban corridors battery eVTOLS cannot serve.

Superior Economics

40% lower OpEx vs helicopters. Profitable at scale with \$140/hour ticket yields.

Market Demand

\$3-4B serviceable niche from operators seeking zero-emission regional solutions.

Execution Progress

Design frozen. Certification consultation active. Pilot ops target: 2026-2027.

SERIES A ASK

\$23.8M

To reach Pilot Operations

USE OF FUNDS

Certification & Testing 35%

Prototype Manufacturing 28%

Pilot Routes Launch 20%

Team & Operations 17%

24 Months Runway

The Problem

Regional & Urban Aviation is Broken & Inefficient

Current solutions for 100–300 km intercity routes and premium urban mobility face fundamental barriers that prevent scalable adoption.

Battery Limits

eVTOLS restricted to 30-60 min, viable only for intra-city hops.

Costly Helicopters

\$170–\$220/hour operating cost. Limited to wealthy elites.

Emission Pressure

Regulatory pressure tightens. 2030+ mandates demand zero-emission alternatives.

Intercity Congestion

100–300km demand outpaces ground capacity. Air gap exists.

THE MARKET GAP: Missing a clean, long-endurance aircraft for intercity (100-300km) & premium urban corridors.

Our Solution

The Pushpak 02 Hydrogen-Electric Hybrid

A clean-sheet design optimized for the "missing middle": 100–300 km intercity routes and premium urban corridors that batteries can't reach and helicopters can't serve cost-effectively.

RANGE

300+ km

ENDURANCE

5+ hours

EMISSIONS

Zero

OPEX SAVINGS

40%

Core Technology

Hybrid Powertrain

Hydrogen fuel cell + battery for peak power

Future-Proof

Zero-emission, aligned with 2030+ mandates

Multi-Role Design

Tourism, medical, government, urban mobility

Cost-Effective Ops

Lower maintenance vs turbine helicopters

Technology Architecture

Hybrid Electric Powertrain

Optimized for safety & endurance, combining hydrogen steady-state efficiency with battery peak power response.

Hydrogen Fuel Cell Stack

Primary cruise power. High-efficiency PEM. Altitude-optimized.

High-Power Battery Pack

Peak load handling. Takeoff/landing power. Emergency reserve.

Dual-Redundancy

Independent power buses. Single-point failure tolerance. Certifiable.

Regenerative Recovery

Energy capture during descent & braking. Extended range.

KEY INSIGHT

132x higher raw energy density than Li-ion batteries.

Hydrogen is the only viable path for zero-emission medium-range aviation.

Operational Breakthroughs

- ✓ Multi-hour, multi-flight ops
- ✓ Rapid refueling (minutes)
- ✓ Global H2 momentum building

Market Opportunity

Intercity Air Mobility | 100-300 KM Range Focus

Our hydrogen-electric platform unlocks specific high-value use cases that battery eVTOLS cannot serve due to range limitations.

Tourism & Charter

Premium eco-tourism and private charter for intercity hops.

Regional Shuttles

Scheduled connections between tier-2 cities and major hubs.

Emergency & Gov

Air ambulance, organ transport, disaster response, border ops.

Logistics

Time-critical cargo and infrastructure inspection missions.

MARKET TAM

\$14–16B

Global UAM + Regional Mobility (2032)

SERVICEABLE NICHE

\$3–4B

Hydrogen Intercity (100–300km)

Business Model

Diversified Revenue Streams for Scale & Stability

Aircraft Sales CORE REVENUE	Direct sales to fleet operators, tourism companies, government agencies. Target: \$15-20M per unit.
Operational Services HIGH MARGIN	Establish & manage intercity shuttle routes through strategic operator partnerships & revenue-sharing.
Maintenance & Support RECURRING	Long-term service contracts, spare parts, predictive maintenance (AI-driven), pilot training.
Infrastructure Partnerships ECOSYSTEM	Hydrogen fueling deployment consulting. Airport infrastructure integration. Supply agreements.

Unit Economics

Operator Profitability Model (Conservative)

Key Operating Assumptions

- 6-8 flight hours per day
- 250 operating days per year
- Hydrogen via supply agreements
- Power-by-hour maintenance model

Per Flight Hour Economics

Profitable at Scale

REVENUE / HOUR

~\$140

Avg ticket @ 80% load

DIRECT OPEX

~\$70-90

Fuel, crew, maint

GROSS MARGIN

30-40%

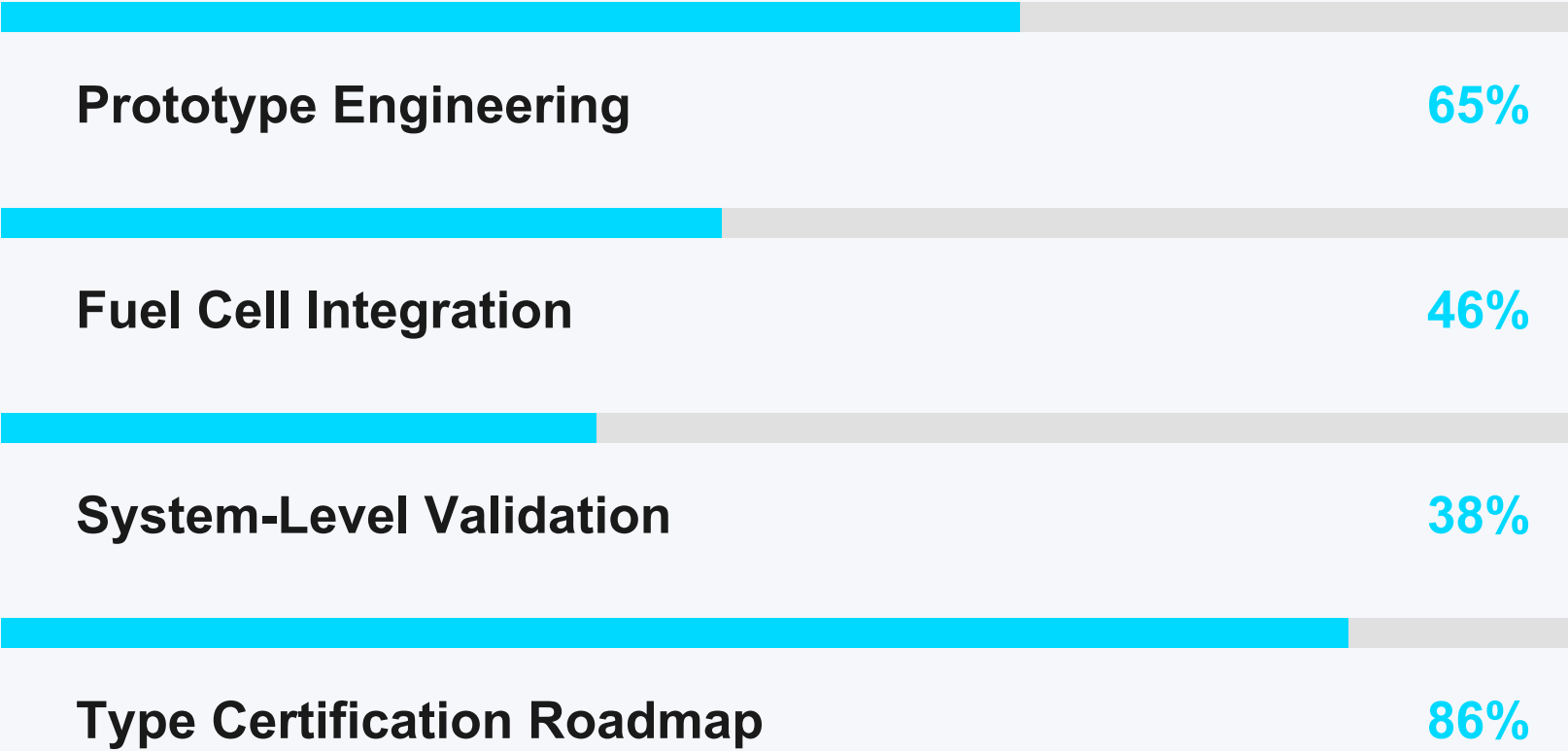
Target range

Development Status

Completed

- ✓ Conceptual Design (full aircraft config locked)
- ✓ Aerodynamic Modeling (preliminary CFD)
- ✓ Hydrogen Safety Framework (protocols established)
- ✓ Certification Consultation (early DGCA/FAA engagement)
- ✓ Preliminary Safety Assessment (risk analysis complete)
- ✓ Supply Chain Mapping (key suppliers identified)

In Progress



Certification Roadmap

Regulatory Path with DGCA + FAA Advisory Support

Phase 1	2024-25	Phase 2	2025-26	Phase 3	2026-27
Foundation & Design ◦ Conceptual Design Review ◦ Preliminary Safety Assessment ◦ Powertrain Bench Testing		Validation & Ground Test ◦ Detailed Design Review ◦ Component Qualification ◦ Ground Test Program		Flight & Certification ◦ Flight Test Campaign ◦ Certification Evaluation ◦ 2027 Type Cert + Ops Approval	

2027 Milestone

Competitive Landscape

Intercity vs. Urban Mobility Solutions

COMPANY	TECHNOLOGY	ENDURANCE	RANGE	MARKET FOCUS
Pushpak 02	Hydrogen-Electric	5+ hrs	300+ km	INTERCITY ROUTES
Joby Aviation	Battery eVTOL	~1 hr	~150 km	Urban
Archer Aviation	Battery eVTOL	~40 min	~60 km	Urban
Helicopters	Jet-A Fuel	2-4 hrs	300-500 km	GENERAL AVIATION

THE PUSHPAK ADVANTAGE

Hydrogen enables longer range and zero emissions for intercity routes — a massive market segment completely underserved by battery eVTOL limitations and priced out by helicopter operations.

Traction & Key Milestones

Commercial Momentum & Strategic Partnerships

Government MOU

SIGNED & ACTIVE

Strategic partnership with state government to establish first 200km pilot corridor for hydrogen aviation.

H2 Infrastructure

IN DISCUSSION

Supply agreements under discussion with major energy providers for airport-side refueling stations.

Supply Chain

PARTNERS IDENTIFIED

Key suppliers locked in for fuel cell stack, high-density batteries, and composite airframe.

Operator Interest

ACTIVE

LOIs from regional charter and tourism operators seeking zero-emission fleet upgrades.

Advisory Board

ASSEMBLED

Former DGCA officials and aerospace powertrain experts onboarded to guide certification.

Go-To-Market Strategy

Phased Rollout: Certification to Global Scale

2025-2027

Prototype & Certification

IN PROGRESS

- Build & validate full-scale prototype
- Secure hydrogen infrastructure partnerships
- Execute flight testing & certification roadmap

2027-2028

Pilot Routes & Market Entry

MARKET ENTRY

- Launch with tourism & charter operators
- Establish regional shuttle networks
- Deploy for emergency & utility missions

2028 Onwards

Scale & Global Expansion

DOMINANCE

- Fulfill multi-aircraft commercial orders
- Full fleet operations & support services
- Expansion to APAC & global markets

⚡ Strategy: Leverage early pilot routes to validate economics before scaling manufacturing capital.

Leadership Team

Deep Tech Engineering Meets Operational Scale



Aneerudh Kumar

FOUNDER & CEO

Vision & Technology Architecture

- ✓ Expert in Hydrogen Fuel Cell Technology
- ✓ Background in Autonomous Flight Systems
- ✓ Aerospace Engineering Pedigree



Vardan Khurana

CO-FOUNDER

Operations & Scale Execution

- ✓ 15+ Years in Operations & Strategy
- ✓ Specialist in Strategic Partnerships
- ✓ Proven Track Record in Venture Scaling

CORE ENGINEERING DOMAINS

 **Hydrogen Systems**

 **Aerodynamics**

 **Avionics**

 **AI & Autonomy**

Series A Ask

Capital Requirement & Allocation Strategy

TOTAL RAISE AMOUNT

\$23.8M

24+ MONTHS RUNWAY

Funds secure operations through critical milestones including prototype completion, ground testing, and regulatory submissions.

PRIMARY OBJECTIVE

Achieving Type Certification readiness and initiating first revenue-generating pilot routes.

To achieve Type Certification and launch	
Use of Funds Allocation	
Certification & Testing	\$8.3M (35%)
Prototype & Manufacturing	\$6.7M (28%)
Pilot Routes & Launch	\$4.8M (20%)
Team & Operations	\$4.0M (17%)

Risks & Mitigation

Proactive Management Strategy

Technology Risk

Hydrogen fuel cell integration in aviation is nascent.

MITIGATION:

- Phased testing (bench → ground → flight)
- Adherence to emerging H2 aviation standards
- Dual-redundancy in critical systems

Certification Risk

Regulatory delays or changing compliance requirements.

MITIGATION:

- Advisory board with former DGCA/FAA officials
- Early & continuous regulator engagement
- Conservative timeline buffers built in

Hydrogen Infrastructure

Limited aviation-grade H2 refueling at regional airports.

MITIGATION:

- Strategic partnerships with energy majors
- Modular/mobile refueling unit deployment
- Point-to-point routes with hub-based fueling

Capital Intensity

High upfront costs for prototype & manufacturing scale.

MITIGATION:

- Staged funding linked to milestones (Series A/B)
- Asset-light model via operator partnerships
- Government grants & non-dilutive sources

Financial Projections

Conservative Growth Trajectory & Path to Profitability

Year	Revenue	EBITDA	Margin %	Stage
2027	\$8-12M	-\$32M	-320%	Prototype & Cert
2028	\$35-50M	-\$18M	-40%	Pilot Routes
2029	\$90-120M	\$5-10M	5-10%	Scaling
2030	\$180-220M	\$40-50M	20-25%	Growth
2031	\$300-350M	\$80-100M	25-30%	Dominance
2032	\$450-500M	\$140-160M	30-35%	Scale

KEY INSIGHTS: EBITDA positive by 2029 | 30%+ margins by 2031 | \$450-500M revenue run-rate by 2032 | Path to \$1B+ valuation

Why Invest Now

Category Emerging

Global UAM market validation happening. Joby, Lilium raising at \$1B+ valuations. Hydrogen aviation gaining momentum.

Regulatory Window

DGCA actively developing certification framework. First-to-certify gains 5+ year competitive moat.

Hydrogen Tailwind

Global hydrogen economy accelerating. Green H2 now cost-competitive. Aviation adoption imminent.

Market Gap Unlocked

No other player pursuing hydrogen intercity VTOL. Clear defensible niche vs crowded battery eVTOL space.

Founder-Market Fit

Aneerudh's aerospace background + Vardan's operations expertise = perfect team for execution.

2027 Inflection Point

First commercial flights = inflection. Early investors → board seats & exits 2032-2035.

Building the Future of Sustainable Flight

Pushpak 02 is building the first hydrogen-electric aircraft in the APAC region designed specifically for intercity mobility — a segment underserved and growing.

CONTACT INFORMATION

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